Time

Units of time: The SI unit of the time is the **second**, written as **s**.



There is always a time lapse between seeing an event and starting (or stopping) use of the stop-watch by hand. This error is called the **reaction time** error.
The reaction time is about 0.2 s for the average person.

The **stop-watch** is used to measure time intervals.

$$\text{Percentage error} = \frac{\text{maximum error}}{\text{measure value}} \times 100\%$$

1. Using a stop-watch, a girl measured the time of fall of a stone from a height. The time recorded was 2.2 s. If the girl's reaction time was 0.2 s, what is the possible percentage error? (2 marks)

Total possible error = 0.2 s + 0.2 s = 0.4 s
∴ Possible percentage error= 0.4 s / 2.2 s × 100% = 18%
Such large percentage error makes the stop-watch reading unacceptable.

Remarks:
∴ stop-watch is not suitable for measuring very short time intervals

2. The figure below shows the instant when athletes in the 100-m sprint are crossing the finishig line. The result of the winner is 10.05 s. If the time is measured by a hand-held stop-watch which has a maximum error of 0.005 s and the overall reaction time of the timekeeper is 0.3 s, what is the maximum possible percentage error of the result? (2 marks)

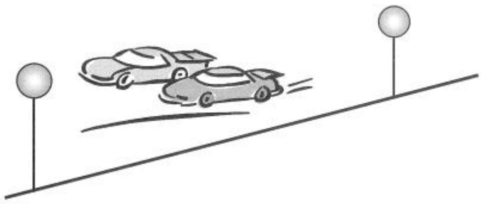
(a) Maximum possible error = 0.3 + 0.005 = 0.305 s

Maximum possible percentage error $= \frac{0.305}{10.05} \times 100\% = 3.03\%$

3. Using a stop-watch, Tom measured the time for a car to pass through 2 streetlights. The reading was 3.2 s. If the reaction time of Tom is 0.2 s, what is the maximum possible percentage error of his measurement?

A 3.1%
B 6.3%
C 12.5%
D 25.0%

maximum % error $= \pm \frac{0.4}{3.2} \times 100\ \% = \pm 12.5\ \%$

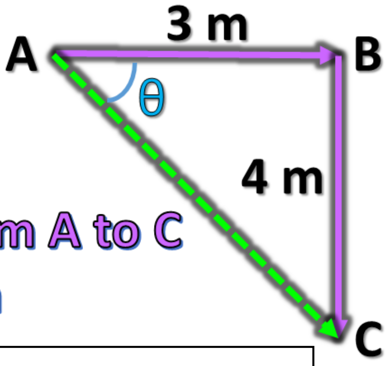


C

Distance and displacement

Unit : meter (m)

Displacement(s) : The length of a **straight line joining** the **starting point** with the **finishing point** in a specified direction



Distance from A to C
= 3 + 4 = 7 m

Displacement from A to C
$= \sqrt{3^2 + 4^2} = 5$ m

$\tan\theta = \frac{4}{3}$

$\theta = 53.1^o = S\ 36.9^o\ E$

Distance(d): The total path length travelled by the object

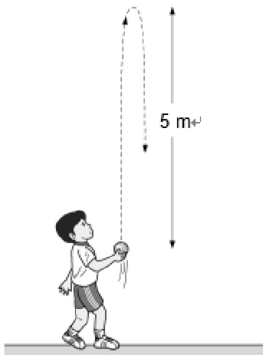
4. A boy throws a ball vertically upwards as shown. What is the displacement of the ball when it returns to its starting position?

   A 0 m

   B 5 m

   C 5 m upwards then 5 m downwards

   D 10 m

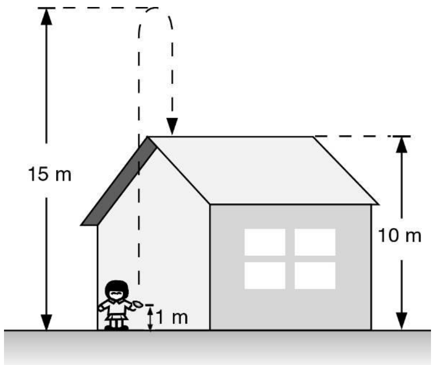


The final position of the ball is the same as the starting position. Therefore, the total displacement is zero.

A

5. The following figure shows a girl throwing a stone vertically upwards. The stone leaves her hand when it is 1 m above the ground. Then it rises to 15 m above the ground and falls down to the roof of a house. The roof of the house is 10 m high. What are the total distance travelled by the stone and its displacement for the whole journey?

	Total distance	Total displacement
A.	19 m	9 m (upward)
B.	20 m	9 m (upward)
C.	20 m	10 m (upward)
D.	15 m	10 m (upward)



Total distance travelled = (15 – 1) + (15 – 10) = 19 m
Displacement = 9 m (upwards)

A

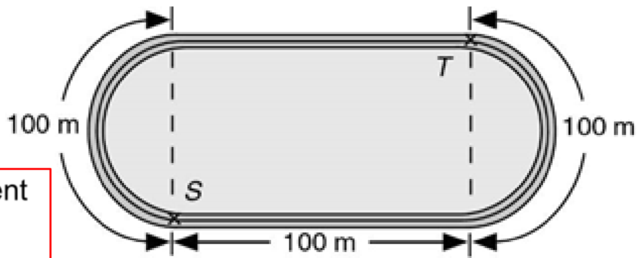
6. In a 1 km run, a student starts from S and ends at T as shown in the following figure. What is the magnitude of his displacement?

   A. 119 m

   B. 200 m

   C. 226 m

   D. 1000 m



Magnitude of his total displacement
$= \frac{100 \times 2}{\pi}$
$= 63.7$ m

A

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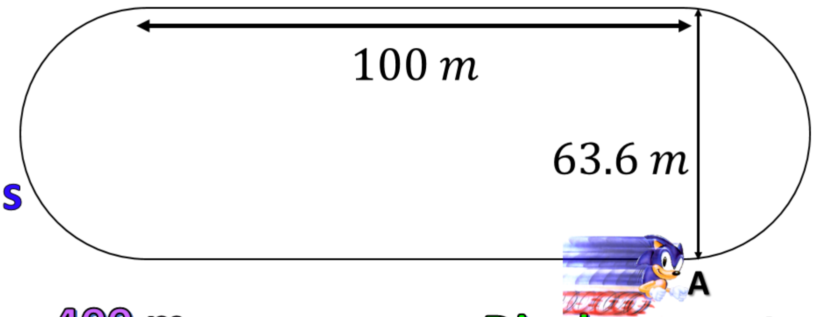
Speed and velocity

Unit = ms^{-1}

Time = 40 s



Distance = 400 m

Displacement = 0 m

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{400\text{ m}}{40\text{ s}} = 10\text{ ms}^{-1}$$

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} = \frac{0\text{ m}}{40\text{ s}} = 0\text{ ms}^{-1}$$

7. A man takes 30 s to walk from point A to point B along a straight road, where AB = 120 m. He then takes 20 s to run backwards to a point C, where BC = 100 m. Find the average speed of the man for the whole journey

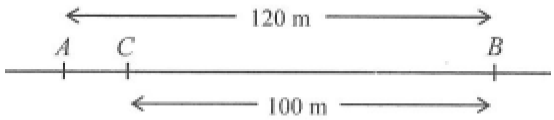
A 2.0 m s^{-1}

B 4.0 m s^{-1}

C 4.4 m s^{-1}

D 4.5 m s^{-1}

$v = \frac{s}{t} = \frac{120+100}{30+20}$ $\therefore v = 4.4\text{ m s}^{-1}$



C

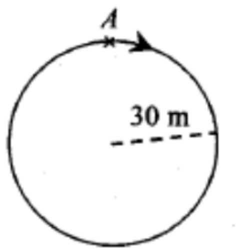
8. A car starts at point A and travels along a circular path of radius 30 m. After 15 s, the car returns to point A. Find the average speed of the car within this period of time.

A. Zero

B. 2 m s^{-1}

C. 6.3 m s^{-1}

D. 12.6 m s^{-1}



D

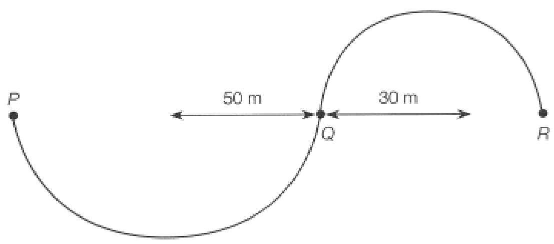
9. Tom walks along a curved path PQR, which is made up of two semi-circular parts PQ and QR of radius 50 m and 30 m respectively. He takes 40 s to walk from P to Q and 15 s to walk from Q to R. The magnitude of his average velocity is

A. 2.91 ms^{-1}

B. 3.25 ms^{-1}

C. 4.57 ms^{-1}

D. 5.0 ms^{-1}



A

10. Tsing Ma Bridge is 2160 m long. If a car moves at a constant speed of 80 km h^{-1}, how long does it take to travel through the whole bridge?

A 7.5 s

B 27 s

C 34.4 s

D 97.2 s

Time taken

$= \frac{2.16}{80}$ h

$= \frac{2.16}{80} \times 60 \times 60$ s

$= 97.2$ s

D

11. A car takes 8 minutes to travel along a path *OPQR* on a horizontal surface as shown. Given that *OP* = *PQ* = 2 km, find the magnitude of the average velocity of the car in this journey.

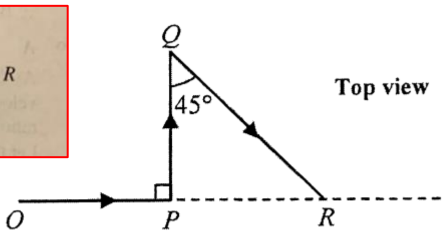
A. 30 kmh^{-1}

B. 36 kmh^{-1}

C. 41 kmh^{-1}

D. 51 kmh^{-1}

Note that velocity = displacement ÷ time, not distance travelled ÷ time.
Magnitude of the total displacement of the car = distance between O and R
$= OP + PR = 2 + 2\tan 45° = 4$ km
Magnitude of the average velocity of the car $= 4 \div (8 \div 60) = 30$ km h^{-1}



A

12. A car travels with constant speed v along a horizontal road *PQRST* with four sections of equal length as shown. Determine v if the car's average velocity for the journey *PQRST* is 20 km h^{-1} in magnitude.

A. 10 km h^{-1}

B. 20 km h^{-1}

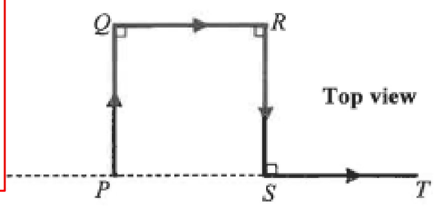
C. 40 km h^{-1}

D. It cannot b

By average speed $= \frac{\text{distance travelled}}{\text{time taken}}$, $v = \frac{PQ + QR + RS + ST}{t}$ ..... (*1)

By average velocity $= \frac{\text{displacement}}{\text{time taken}}$, $20 = \frac{PT}{t}$ .... (*2)

(*1) ÷ (*2): $\frac{v}{20} = \frac{PQ + QR + RS + ST}{PT} = \frac{4}{2} \Rightarrow v = 40 \text{ kmh}^{-1}$



C

13. A particle travels at 2.0 ms^{-1} due east for 1.5 s and then travels at 4.0 ms^{-1} due north for 1.0 s. What is the magnitude of its average velocity for the whole journey?

A. 2.0 ms^{-1}

B. 2.8 ms^{-1}

C. 3.0 ms^{-1}

D. 5.0 ms^{-1}

Displacement of the whole journey : $s = \sqrt{(2.0\times1.5)^2 + (4.0\times1.0)^2} = 5$ m

Average velocity $= \frac{s}{t} = \frac{(5)}{(1.5+1.0)} = 2.0 \text{ m s}^{-1}$

A

14. Peter walks along a straight road from point P to point Q with an average speed of 2 ms^{-1}. He then runs back from Q to P along the same road with average speed of 4 ms^{-1}.Which of the following statements are correct?

(1) The resultant displacement of Peter in the whole journey is zero

(2) The average velocity of Peter in the whole journey is 0 ms^{-1}

(3) The average speed of Peter in the whole journey is 3 ms^{-1}

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

A

× (3) Assume the distance between P and Q is D.

Time taken to travel from P to Q $= \frac{D}{2}$

Time taken to travel from Q to P $= \frac{D}{4}$

Average speed in whole journey $= \frac{D+D}{D/2 + D/4} = 2.67 \text{ m s}^{-1}$

✓ (1) As Peter returns to the original position, the resultant displacement is zero.

✓ (2) Average velocity is defined as the total displacement per total time taken. As the total displacement is zero, the average velocity must be zero.

15. A boy ran from A to B with a uniform speed v. Then he ran back from B to A with uniform speed 2v. Find his average speed during the whole running

A. $\frac{3v}{2}$ B. $\frac{4v}{3}$

C. $\frac{5v}{4}$ D. $\frac{v}{2}$

Let the distance be d.

$t_1 = \frac{d}{v}$ and $t_2 = \frac{d}{2v}$

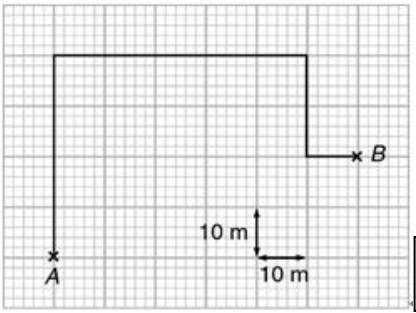
Average speed $= \frac{2d}{t_1 + t_2} = \frac{2d}{d/v + d/2v}$

$= \frac{2d}{3d/2v} = \frac{4v}{3}$

B

16. Stephen takes the following path and walks for 3 minutes from A to B. What are the average speed and the magnitude of the average velocity of Stephen?

	Average speed	Average velocity
A	$0.667\ \text{m s}^{-1}$	$0.441\ \text{m s}^{-1}$
B	$0.667\ \text{m s}^{-1}$	$0.351\ \text{m s}^{-1}$
C	$0.441\ \text{m s}^{-1}$	$0.667\ \text{m s}^{-1}$
D	$0.441\ \text{m s}^{-1}$	$0.441\ \text{m s}^{-1}$



B

Distance travelled = 40 + 50 + 20 + 10 = 120 m

Average speed = $\frac{120}{60\times3}$ = 0.667 m s^{-1}

Magnitude of displacement = $\sqrt{60^2 + 20^2}$ = 63.2 m

Magnitude of average velocity = $\frac{63.2}{60\times3}$ = 0.351 m s^{-1}

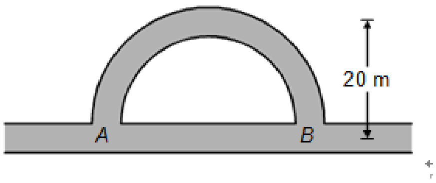
17. The figure below shows two paths connecting points A and B. Mary and Jane travel from A to B along the semicircular path and the straight path respectively. The radius of the semicircular path is 20 m. If Mary and Jane uses the same time of 40 s for their journeys, find the difference in the average speeds of the girls.

A. $0.571\ \text{m s}^{-1}$

B. $1\ \text{m s}^{-1}$

C. $1.57\ \text{m s}^{-1}$

D. $2.14\ \text{m s}^{-1}$



A

18. The figure above shows three paths P_1, P_2 and P_3 from X to Y on a horizontal plane. Three students take the same time to travel from X to Y via the three paths respectively. Which of the following physical quantities about their journey is/are the same?

(1) Displacement

(2) Distance

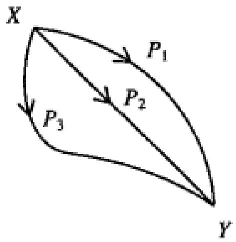
(3) Average speed

A. (1) only

B. (2) only

C. (1) and (3) only

D. (2) and (3) only



(1) ✓ Displacement is defined as the shortest distance between two points. Thus the displacement of a journey only depends on the locations of the starting point and the ending point.

Since the three paths P_1, P_2 and P_3 have the same starting point (i.e. X) and the ending point (i.e. Y), the displacements of the three journeys are the same.

(2) ✗ Distance is defined as the length of the path.

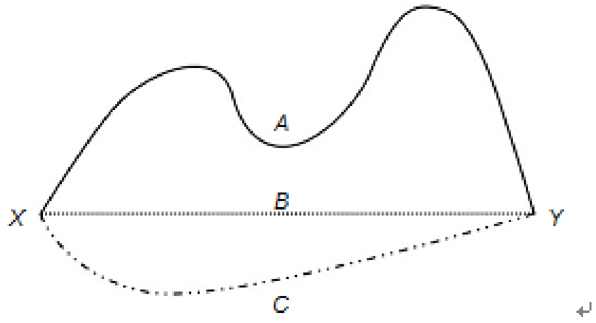
Since the three paths P_1, P_2 and P_3 have different lengths, the distances of the three journeys are different.

(3) ✗ Average speed $= \frac{\text{Distance travelled}}{\text{Time of travel}}$

Along the three paths P_1, P_2 and P_3, the distances travelled are different (as shown in (2)) but the times of travel are the same, thus the average speeds are different.

A

19. Three girls A, B and C run at steady speeds from point X to point Y along different paths as shown. They take the same amount of time to complete the journey.
Which of the following statements are correct?
(1) The girls travel with constant velocity.
(2) The girls have the same average velocity.
(3) Girl A has the highest speed.
A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)



C

Scalar and vector quantities

- **Scalar** is a quantity which is has **magnitude** (size) only but no direction.
  Example of **scalar:** time, mass, temperature, length, distance, speed, etc.
- **Vector** is a quantity which has **both magnitude and direction**.
  Example of **vector:** displacement, velocity, acceleration, force, momentum, etc

20. Which of the following quantities is a vector?
A. Temperature
B. Displacement
C. Distance
D. Speed

B

21. Which of the following is/are **incorrect**?
(1) Distance is a vector which has both magnitude and direction.
(2) Displacement is a vector which has both magnitude and direction.
(3) Speed is a scalar which has magnitude only.
(4) Time is a vector which has both magnitude and direction.
A (1) only
B (1) and (4) only
C (1), (2) and (3) only
D None of the above

B

22. Which of the following statements is **incorrect** about scalars and vectors?
A Scalar quantities have magnitude only and no direction.
B Vector quantities must be specified by magnitude and direction.
C Two vectors in different directions can be added to form a new vector.
D Two vectors are identical if they have the same direction.

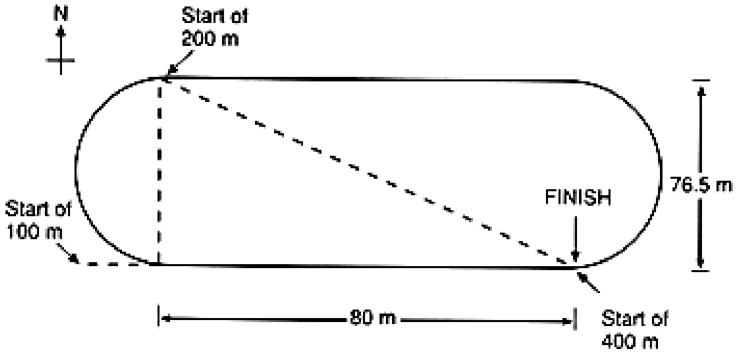
D

23. The women's world record for 200 meter sprint is 21.34 s which is held by Joyner of the United States. The shape of the track is shown in the diagram below. Find

(a) her average speed; and

(b) her average velocity.

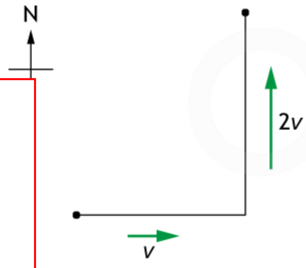
(a) 9.37ms^{-1}
(b) 5.2 ms^{-1} S46.3°E



24. A man walks at velocity v towards the east for time t. Then he walks at velocity $2v$ towards the north for time $2t$ (Fig g). His average speed over the whole journey is 2 m s^{-1}.

(a) Find the direction of his average velocity over the whole journey. (2 marks)

(b) Find the magnitude of his average velocity over the whole journey. (3 marks)



(a) Let θ be the angle between the north and the displacement.

$\tan\theta = \dfrac{vt}{2v \times 2t} = \dfrac{1}{4}$ 1M

$\theta = 14.0°$

The direction of his average velocity is N14.0°

1A

(b) Average speed $= \dfrac{\text{total distance travelled}}{\text{total time of travel}}$

$2 = \dfrac{vt + 2v \times 2t}{t + 2t}$ 1M

$2 = \dfrac{5vt}{3t}$

$v = 1.2 \text{ m s}^{-1}$

Magnitude of his average velocity

$= \dfrac{\sqrt{(vt)^2 + (2v \times 2t)^2}}{t + 2t}$ 1M

$= \dfrac{\sqrt{5v^2t^2}}{3t}$

$= \dfrac{\sqrt{5v^2}}{3}$

$= \dfrac{\sqrt{5 \times 1.2^2}}{3}$

$= 0.894 \text{ m s}^{-1}$ 1A

25. Two cars, P and Q, are moving in the same direction with uniform speeds of 6.0 ms^{-1} and 10 ms^{-1} respectively. At t = 0, P is 10 m ahead of Q. How long will Q take over P.

2.5 s

26. Kitty swims across a 50-m swimming pool from A to D while her coach takes her time beside the pool. The coach starts timing at A. Then he walks along the edge of the pool through B and C to D. The coach and Kitty reach D at the same time and the coach stops timing at this instant. The time recorded is 1 min 15 s.

(a) Find Kitty's average speed in moving from A to D. (2 marks)

(b) Find the coach's average speed in moving from A to D. (1 mark)

(c) Tom says that the coach and Kitty have the same average velocity in moving from A to D. Comment on his statement. (2 marks)

(a) Kitty's average speed $= \dfrac{\text{total distance travelled}}{\text{total time of travel}}$

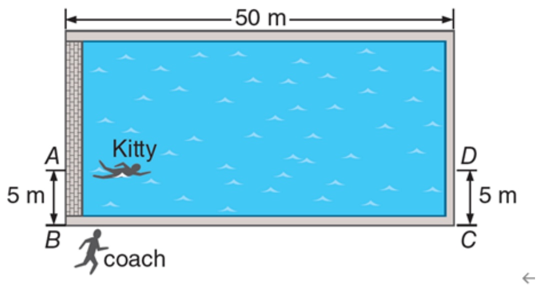
$= \dfrac{50}{1\times 60+15}$

$= 0.667 \text{ m s}^{-1}$

(b) The coach's average speed $= \dfrac{5+50+5}{1\times 60+15} = 0.8 \text{ m s}^{-1}$

(c) Tom is correct.

The coach and Kitty have the same displacement within the same period of time.



27. A car travels 1500 m forwards and then 3000 m backwards.

(a) What is the total distance travelled by the car? (1 mark)

(b) Take the forward direction as positive.

(i) Find the total displacement of the car. (2 marks)

(ii) The whole trip takes 6 minutes. Find the average speed and average velocity of the car. Give your answers in units of km h^{-1}. (4 marks)

(a) Total distance travelled by the car = 1500 + 3000 = 4500 m

(b) (i) Total displacement of the car = 1500 + (−3000)

= −1500 m

(ii) Average speed $= \dfrac{\text{total distance travelled}}{\text{total time of travel}}$

$= \dfrac{4500 \div 100}{6 \div 60}$

$= 45 \text{ km h}^{-1}$

Average velocity $= \dfrac{\text{total displacement}}{\text{total time of travel}}$

$= \dfrac{-1500 \div 1000}{6 \div 60}$

$= -15 \text{ km h}^{-1}$

28. A jumbo jet flies at 260 m s^{-1} towards the north for 10 s. It then flies at 254 m s^{-1} towards the west for 8 s.

(a) Find the total distance it has travelled. (2 marks)

(b) Find its average speed in the whole journey. (1 mark)

(c) Find the magnitude of its average velocity in the whole journey. (3 marks)

(a) By average speed $= \dfrac{\text{total distance travelled}}{\text{total time of travel}}$,

total distance travelled = 260 × 10 + 254 × 8 = 4630 m

(b) Average speed in the whole journey $= \dfrac{4630}{10+8} = 257 \text{ m s}^{-1}$

(c) Magnitude of total displacement $= \sqrt{(260\times 10)^2 + (254\times 8)^2} = 3300$ m

Magnitude of average velocity $= \dfrac{\text{total displacement}}{\text{total time of travel}}$

$= \dfrac{3300}{10+8}$

$= 183 \text{ m s}^{-1}$